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SUMMARY

Computational physicist and research engineer with 10+ years building research software for physics at scale: the simulation infrastructure, tooling, and agent systems that scientific ML runs on, the large-scale ground truth those models train against, and contributions to the model architectures themselves.

- **Simulation at scale:** multi-component, interface-resolved campaigns at **140M-cell / 7,000-core / ~2.8M CPU-hour** scale on HPC clusters, validated head-to-head against flyer-impact experiments at UIUC and LANL; now leading the Iowa-side effort, with IIT Madras and Iowa State, to take a five-institution AMReX port to seamless GPU execution and scaling on lab-specific reactive multi-material problems (**ATPESC 2026**).
- **Physics-aware machine learning:** made the case for the deformable-convolution formulation in D-PARC, where 19.7M parameters beat a 243.5M-parameter PARC-family baseline; build pipelines feeding diffusion-, U-Net- and CycleGAN-generated microstructures into physics solvers; define conservation, interface and entropy criteria a surrogate must pass.
- **Research software and agent systems:** independently built and open-sourced a permission-scoped container whose rule-generating program emits a 586-rule permission engine, letting autonomous coding agents work on IP-restricted research code without reading the restricted parts; plus multi-agent orchestration and reproducible build pipelines in daily use.

CORE SKILLS

- **Machine learning:** physics-aware neural operators (PARC, D-PARC); generative models for microstructure synthesis (diffusion models, U-Net, CycleGAN) feeding physics solvers; surrogate modeling and physics-consistency evaluation (conservation/interface/entropy conditions); **PyTorch**; *adjacent/conversant*: PINNs, FNO, differentiable simulation.
- **Research software & systems:** Python, C++, Fortran90, Bash; MPI, domain decomposition, parallel I/O, Slurm/PBS; multi-thousand-core HPC campaigns; post-processing, QoI extraction and visualization of large simulation fields; GPU portability across CUDA / SYCL / HIP (**ATPESC 2026**); profiling and debugging at scale; Git, LaTeX, Jupyter; reproducible build pipelines, provenance / lineage tracking; multi-agent orchestration; permission-scoped agentic-AI development environments.
- **Scientific computing & materials:** sharp-interface and ghost-fluid multi-material methods; finite-volume, high-order shock capturing; level-set representation of microstructural heterogeneity; **AMReX** block-structured AMR; constitutive models and equations of state; composite microstructures from SEM and μ -CT imagery.

EXPERIENCE

Postdoctoral Research Scholar — IIHR–Hydroscience & Engineering, University of Iowa

May 2025 – Present · Iowa City, IA

- Made the architectural case for deformable sampling in **D-PARC**, a physics-aware neural operator in the PARC family: a recurrent differentiator-integrator pair that learns the differential and integral operators of the governing PDE; at **19.7M parameters** it beat a **243.5M-parameter** variant from the same family on reactive-flow prediction (*under review in APL Machine Learning*; preprint at [arXiv:2601.11657](https://arxiv.org/abs/2601.11657)).
- Designing an agentic, closed-loop materials-discovery framework with generative inverse design and leading its microstructure-to-prediction sub-pipeline: routing real (SEM, μ -CT) and ML-generated micrographs to physics-based (**SCIMITAR3D**) or neural (**PARC-family**) predictors, and owning where automation enters the loop and how data and provenance move between simulation and experimental stores; leading 1 postdoc and 2 PhD students.
- Simulation-side lead for University of Virginia's ML group ([Prof. Baek](#)): set coverage targets over the shock-loading parameter space, run the DNS campaigns that meet them, and define the pass/fail benchmarks – peak pressure, temperature localization, reaction-front velocity – against which surrogates are accepted or rejected. Results feed a newly awarded **ONR-MURI**.
- Contributing numerical-method concepts from **SCIMITAR3D** into the group's next-generation **AMReX**-based multiphysics solver through sustained design review; since **ATPESC 2026**, leading the Iowa-side group — with IIT Madras and Iowa State — that tests seamless GPU operation and scaling of the ported baseline for high-speed multi-material reactive dynamics across CUDA, SYCL and HIP backends.

Graduate Research Assistant (PhD) — University of Iowa

Aug 2019 – Apr 2025 · Iowa City, IA

- Extended **SCIMITAR3D** (in-house hydrocode) into a multi-material sharp-interface framework resolving composite microstructures as discrete level-set entities; enforced the consistency conditions making the Riemann-based ghost-fluid method compatible with 5th-order WENO reconstruction, which reaches the same error with **3.3× fewer cells in 3D** than 3rd-order ENO ([Okafor et al., Shock Waves, 2026](#)).

- Designed and executed production-scale HPC campaigns on DoD HPC clusters; **140M-cell grids, 7,000 CPU cores, ~2.8M CPU-hours at 15 nm cell-size** over $50 \times 70 \mu\text{m}$ mesoscale sub-samples (*published in Shock Waves, 2026*).
- Validated simulated predictions head-to-head against experimental data from UIUC (Prof. Dlott) and LANL, selecting the constitutive models and grid criteria that make sub-grain comparison meaningful; coordinated through a seven-university **AFSR-MURI** (*first-authored in J. Applied Physics, 2022*).
- Led simulation and physics-validation for **HEDS**: a deep-learning + computational-mechanics pipeline learning microstructure-to-sensitivity mappings; contributed the evaluation criteria, solver integration and interpretation of learned representations (*published in APL Machine Learning, 2025*).

Project Associate — Center for Propulsion Technologies, IIT Madras, India

Jul 2018 – Jun 2019 · Chennai, India

- Contributed to DRDO's **Variable Camber Morphing Wing** project through immersed-boundary implementation and aerodynamic analysis.

Graduate Research Scholar (MS) — IIT Madras, India

Jul 2015 – Jun 2018 · Chennai, India

- Characterized normal-shock / boundary-layer interaction and porous-medium shock control through coupled finite-volume RANS campaigns and transonic wind-tunnel experiments (*Shock Waves 2021; AIAA Journal 2019*).

SELECTED PUBLICATIONS

Full list with 7 journals (first author on 5), 1 preprint, 4 proceedings, 13 presentations at [Google Scholar](#).

1. Beerman, J.T., **Roy, S.**, Udaykumar, H.S., Baek, S.S.: *Size is Not the Solution: Deformable Convolutions for Effective Physics-Aware Deep Learning*, arXiv:2601.11657 (2026) — under review, **APL Machine Learning**.
2. **Roy, S.**, Seshadri, P.K., Okafor, C., Johnson, B.P., & Udaykumar, H.S.: *High-Fidelity Simulations of Shock Initiation of an Energetic Crystal-Binder System Due to Flyer Impact*, Shock Waves 36(2), 2026.
3. Okafor, C., Seshadri, P.K., **Roy, S.**, & Udaykumar, H.S.: *High-order Eulerian Sharp Interface Numerical Techniques for Micro- to Macro-Scale Simulation of Shocked Energetic Materials*, Shock Waves 36, article 18, 2026.
4. Fang, I., **Roy, S.**, Nguyen, P.C.H., Baek, S.S., Udaykumar, H.S.: *Heterogeneous Energetic Material Damage Simulator (HEDS): A Deep Learning Approach to Simulate Damage-Sensitivity Linkages*, APL Machine Learning 3(2), 2025.

OPEN SOURCE & TECHNICAL WRITING

- [claude-docker-sandbox](#) – containment for running an autonomous coding agent on IP-sensitive research code: a rule-generating program emits the 586-rule permission engine that, with OS-enforced container limits and an agent behavioral contract, keeps restricted source compilable by the build system but unreadable by the agent. Validated by decoy test and in daily use on my own research code; written up as a two-part series in *Level Up Coding* ([part 1](#), [part 2](#)).
- [89% of my coding-agent bill was cache traffic](#) – a measurement study of 3.44 billion tokens of my own Claude Code transcripts, and why most token-saving tools cannot reach that spend (*Level Up Coding*, 2026).

EDUCATION

- **PhD, Mechanical Engineering** — University of Iowa, 2025 · Advisor: [Prof. H.S. Udaykumar](#). *Dissertation: A versatile framework for high-fidelity computations of high-speed reactive multi-material dynamics*.
- **MS (Research), Aerospace Engineering** — IIT Madras, India, Jul 2015 – Jun 2019 · Advisors: [Prof. Santanu Ghosh](#) & [Prof. G. Rajesh](#). Thesis: Investigation of drag reduction using a porous medium for a normal-shock/boundary-layer interaction.
- **BTech, Mechanical Engineering** — NIT Durgapur, India, 2015.

AWARDS, SERVICE & TEACHING

- **Argonne Leadership Computing Facility** (2026) — Director's Discretionary Allocation, Principal Investigator (5,000 Aurora + 1,000 Polaris node-hours); selected participant, Argonne Training Program on Extreme-Scale Computing ([ATPESC](#)), fully funded.
- **Ballard & Seashore Dissertation Fellowship** (2025) and **Dean's Graduate Fellowship**, College of Engineering (2019) — University of Iowa.
- **Finalist (Top 13)**, University of Iowa Graduate College Three Minute Thesis (3MT), selected through a graduate-college-wide preliminary round (2024).
- **Excellent Presentation Award**, 7th International Symposium on Energetic Materials and Their Applications, Tokyo, Japan (2021).
- **Peer Reviewer**: *Journal of Applied Physics* (AIP) · *Physics of Fluids* (AIP) · *AIAA Journal* · *Mechanics of Materials*.
- **Teaching Assistant** (U Iowa side), *Physics-Aware Deep Learning — Conservation laws for physical systems, PINNs, neural operators, PARC, reduced-order models*, offered jointly with the UVA School of Data Science (Spring 2026); further graduate coursework in Deep Learning and Statistical Learning.